Problem Set

Veronica Pletsch

2026-01-21

Contents

Analysis; Is War in Decline? **6**

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(readr)
```

```r
cap <- read.csv("cap.csv")
wars <- read.csv("wars.csv")
```

```r
#merge wars and cap file with inner_join by matching ccode and year/startyear1
merged_wars <- wars %>%
  inner_join(cap, by = c("ccode" = "ccode", "StartYear1" = "year")) %>%

#convert tpop to thousands so that it matches batdeath
mutate(population_total = tpop * 1000) %>%

#collapse multiple rows for one war into 1 row with totals
group_by(WarName, StartYear1) %>%

#aggregate deaths
summarize(
    total_deaths = sum(BatDeath, na.rm = TRUE),
    total_belligerent_pop = sum(population_total, na.rm = TRUE),
    .groups = "drop"
  ) %>%
```

```r
#calculate the total deaths for belligerents
mutate(
    deaths_by_poptotal = total_deaths / total_belligerent_pop
  )
```

```r
#rename year variable
merged_wars <- merged_wars %>%
  rename(year = StartYear1
         )
```

```r
#save merged_wars as new csv file
write.csv(merged_wars, "merged_wars.csv")
```

```r
#Plot a histogram of your deaths per capita distribution and make sure you see the long tail
library(tidyverse)
```

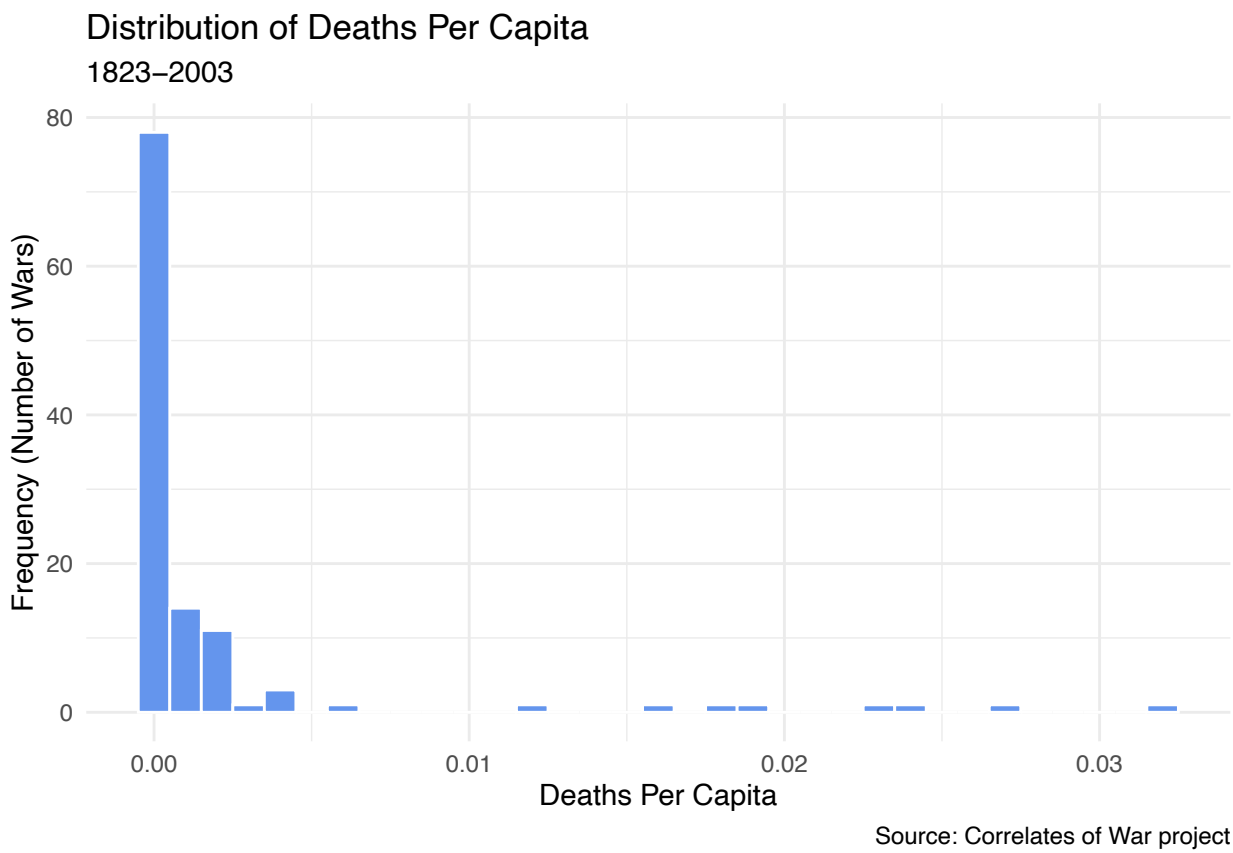
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

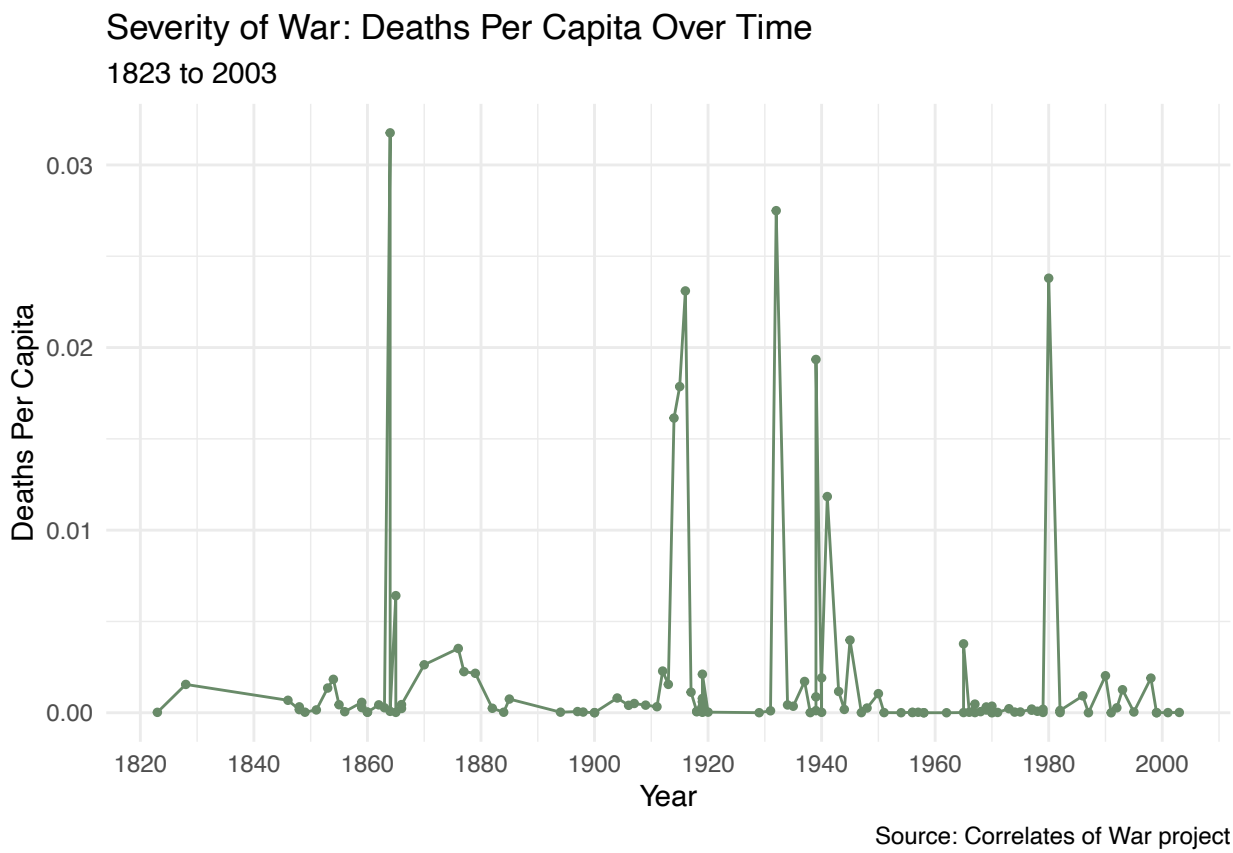
```r
ggplot(merged_wars, aes(x = deaths_by_poptotal)) +
  #changed bins and binwidth to capture the longtail
  geom_histogram(fill = "cornflowerblue", color = "white", bin = 100, binwidth = 0.001) +
  labs(
    x = "Deaths Per Capita",
    y = "Frequency (Number of Wars)",
    title = "Distribution of Deaths Per Capita",
    subtitle = "1823-2003",
    caption = "Source: Correlates of War project"
  ) +
  theme_minimal()
```

```
## Warning in geom_histogram(fill = "cornflowerblue", color = "white", bin = 100,
## : Ignoring unknown parameters: `bin`
```




```r
#Plot deaths per capita against the year, and see if you notice a pattern.

library(tidyverse)
ggplot(merged_wars, aes(x = year, y = deaths_by_poptotal)) +
  geom_line(color = "darkseagreen4") +
  geom_point(color = "darkseagreen4", size = 1) +
    scale_x_continuous(breaks = seq(1820, 2000, by = 20))+
  labs(
    title = "Severity of War: Deaths Per Capita Over Time",
    subtitle = "1823 to 2003",
    x = "Year",
    y = "Deaths Per Capita",
    caption = "Source: Correlates of War project"
  ) +
  theme_minimal()
```




```
#install.packages("poweRlaw")

library(poweRlaw)

wars_before_1945 <- merged_wars %>% filter(year < 1945)

wars_after_1945 <- merged_wars %>% filter(year >= 1945, deaths_by_poptotal > 0)

#Before 1945
#Never used a power law before so I am relying on google and chatgpt to guide me
#Create a continuous power-law object
before_45 <- conpl$new(wars_before_1945$deaths_by_poptotal)
#Estimate $x_{min}$ (the value where the power law starts)
est_before_45 <- estimate_xmin(before_45)
before_45$setXmin(est_before_45)
#Estimate $\alpha$ (the scaling parameter)
alpha_before_45 <- before_45$pars


#After 1945
#Create a continuous power-law object
after_45 <- conpl$new(wars_after_1945$deaths_by_poptotal)
#Estimate $x_{min}$ (the value where the power law starts)
est_after_45 <- estimate_xmin(after_45)
after_45$setXmin(est_after_45)
#Estimate $\alpha$ (the scaling parameter)
```

```r
alpha_after_45 <- after_45$pars

#results
cat("Alpha before 1945:", alpha_before_45, "\n")
```

```
## Alpha before 1945: 1.567258
```

```r
cat("Alpha after 1945:", alpha_after_45, "\n")
```

```
## Alpha after 1945: 1.616818
```

```r
#1000 times
alpha_differences <- numeric(1000)

#use a for loop to get to 1000 simulations
for (i in 1:1000) {

#bootstrp sample
sample_before_45 <- sample(wars_before_1945$deaths_by_poptotal, replace = TRUE)
sample_after_45 <- sample(wars_after_1945$deaths_by_poptotal, replace = TRUE)

#estimate the powerlaw for before 1945 sample
before_45_simulation <- conpl$new(sample_before_45)
before_45_simulation$setXmin(estimate_xmin(before_45_simulation))

#estimate the powerlaw for after 1945 sample
after_45_simulation <- conpl$new(sample_after_45)
after_45_simulation$setXmin(estimate_xmin(after_45_simulation))

#Note the difference between before and after 1945
alpha_differences[i] <- before_45_simulation$pars - after_45_simulation$pars

}
```

```r
#Calculate the 2.5% and 97.5% quantiles
conf_interval <- quantile(alpha_differences, probs = c(0.025, 0.975))

#results
print(conf_interval)
```

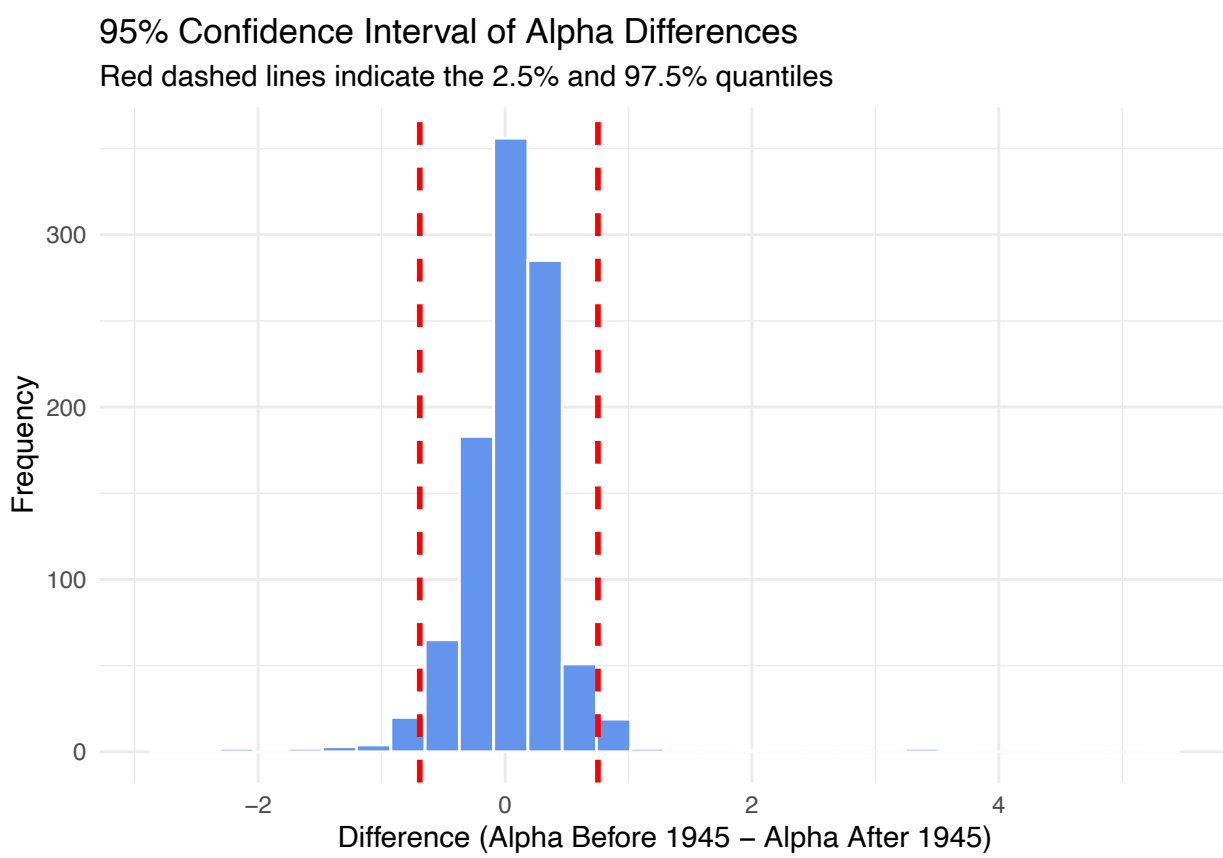
```
##       2.5%      97.5%
## -0.6908720  0.7523444
```

```r
library(tidyverse)

simulated_parameters <- data.frame(diff = alpha_differences)
ggplot(simulated_parameters, aes(x = diff)) +
  geom_histogram(fill = "cornflowerblue", color = "white", bins = 30) +
  geom_vline(xintercept = conf_interval[1], linetype = "dashed", color = "red", size = 1) +
  geom_vline(xintercept = conf_interval[2], linetype = "dashed", color = "red", size = 1) +
  labs(
```

```
    title = "95% Confidence Interval of Alpha Differences",
    subtitle = "Red dashed lines indicate the 2.5% and 97.5% quantiles",
    x = "Difference (Alpha Before 1945 - Alpha After 1945)",
    y = "Frequency"
  ) +
  theme_minimal()
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



Analysis; Is War in Decline?

Based on my analysis, I can say definitively that war is not in decline. Yet, whether war is more or less deadly post 1945 is still up for interpretation. According to my "Severity of War: Deaths Per Capita Over Time" graph, I observe that the frequency of wars has increased. Indeed, between 1823 and 1945 (122 years) there were 69 wars, in comparison to 1945 and 2003 (58 years) where there were 45 wars. That is a ratio of 0.57 wars per year before 1945 versus 0.78 wars per year after 1945, an obvious increase. However, despite this apparent difference, the alpha parameter from the "powerlaw" test shows that there are less deaths per capita per war after 1945 (a=1.616818) in comparison to before 1945 (a=1.567258). Moreover, based on the 95% confidence interval, there is no statistical significance in deaths per capita before versus after 1945.

Regardless of these statistical outcomes, the data is incomplete and an analysis of war data between 1945 and 2023 (equivalent to the 122 years of data between 1823 and 1945) is required for an accurate comparison and definitive observation of whether war is more or less deadly post 1945.